5 **(a)** Monochromatic light is incident on four rectangular slits as shown in Fig. 5.1.

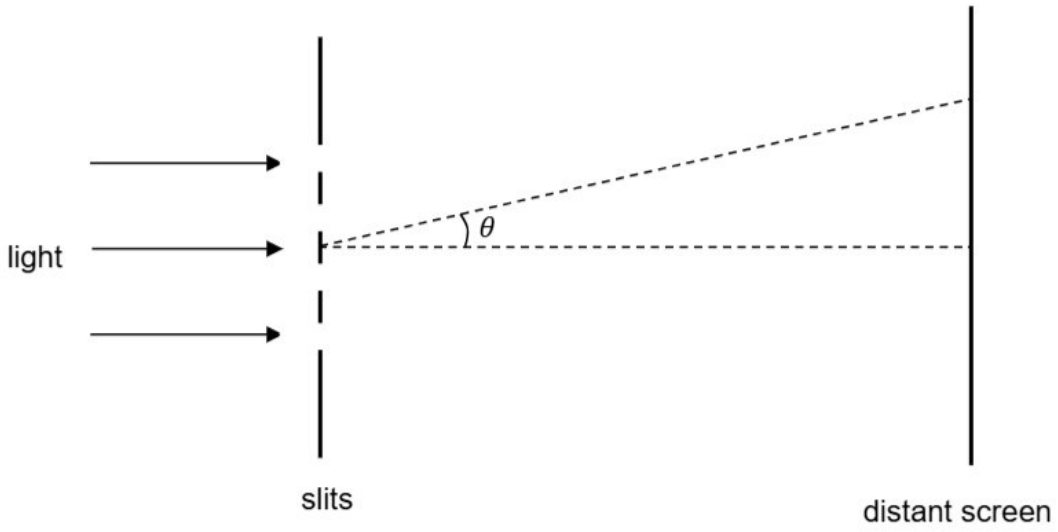


Fig. 5.1 (not to scale)

The graph in Fig. 5.2 shows the variation with angle θ of the intensity of the light I at the distant screen.

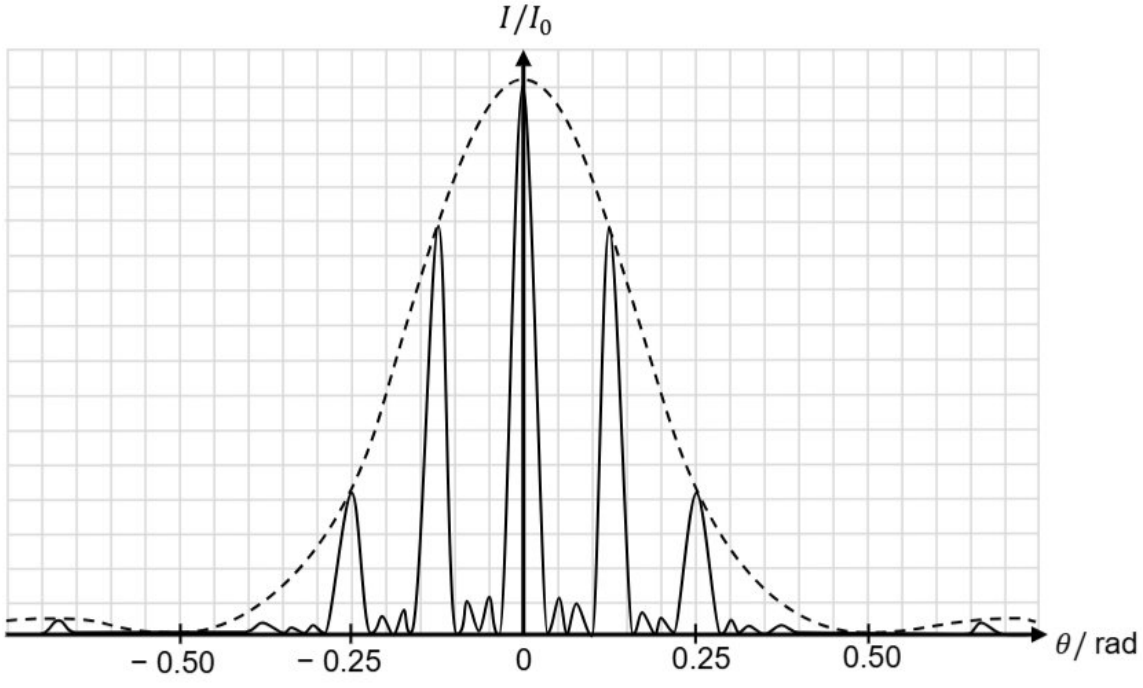


Fig. 5.2

(i) Explain the significance of the dotted curve on the graph.

.....

.....

..... [1]

(ii) The width of each slit is 1.0 μm.
Use Fig. 5.2 to show that the wavelength of the light is 450 nm.

[1]

(iii) The distance to the screen is 3.2 m.
Determine the separation between the slits.

separation = ... m [2]

(iv) The number of slits is increased while keeping the width of each slit and the separation of slits the same.

State **two** changes to the maxima observed on the screen as a result of the increased number of slits.

.....

.....

.....

..... [2]

(b) A tube is filled with water and a speaker is placed above the tube, playing a constant 310 Hz tone, as shown in Fig. 5.3 below.

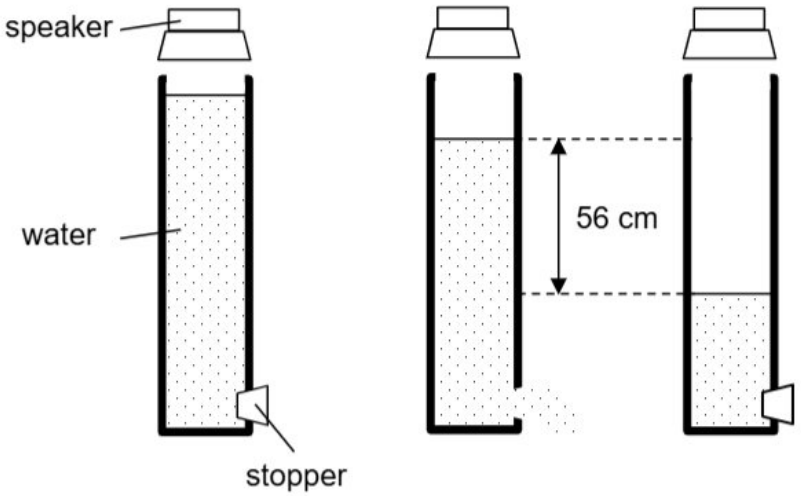


Fig. 5.3 (not to scale)

The stopper is removed and water is allowed to run out of the tube. At the position shown in the centre diagram in Fig. 5.3, a loud sound is heard in the tube for the first time.

Water continues to run out of the tube, and a loud sound is next heard at the position shown in the rightmost diagram in Fig. 5.3.

The stopper is re-inserted to maintain the water level at this position.

(i) On the rightmost diagram of Fig. 5.3, sketch the standing wave in the tube and identify with the letter 'N', the positions of any nodes of the standing wave.

[2]

(ii) Calculate the speed of sound in the tube.

speed of sound = ... m s^{-1} [2]

(iii) With the water level maintained as shown in the rightmost diagram of Fig. 5.3, the frequency of the sound played by the speaker is increased from 310 Hz until the next loud sound is heard.

Calculate the new frequency of the sound.

frequency = ... Hz [3]